SHARP FAVARD DECAY WITH UNIFORM MARGINALS

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Abstract. We construct a compact planar set of length one whose length measure has uniform coordinate marginals and whose δ-neighborhoods have Favard length of order $\frac{1}{\log \frac{1}{\delta}}$. Hall's matching theorem then gives the sharp logarithmic bound for unions of permutation strips.

Divide the top and bottom edges of the unit square into n equal intervals, match them by a permutation, and join matched intervals by parallelograms. Their areas add to 1; the question is how small their union can be. Writing α_n for the minimum, we call the assertion $\alpha_n \asymp \frac{1}{\log n}$ *KC2.*[1]

A strip joining I to J is encoded by the square $I \times J$ in the plane of endpoints. At height t, its section is the projection $\pi_t(I \times J)$, where $\pi_t(x, y) = (1 - t)x + ty$. A set with a unit coordinate projection has Hausdorff length at least 1. We attain this minimum with both coordinate projections equal to $[0, 1]$, together with a logarithmic bound on the averaged projections of its neighborhoods. For compact $S \subset \mathbb{R}^2$, put $S_\delta = S + [-\delta, \delta]^2$ and define its unnormalized Favard length by

$$\mathrm{Fav}(S) = \int_0^\pi |p_\theta(S)|\, d\theta, \qquad p_\theta(x, y) = x \cos\theta + y \sin\theta.$$

Let p_x, p_y be the coordinate projections. We write $|\cdot|$ for length or area, and $\mathcal{H}^1$ for Hausdorff length.

Theorem 1. *There is a compact set $S \subset [0, 1]^2$ such that*

$$p_x(S) = p_y(S) = [0, 1], \qquad \mathcal{H}^1(S) = 1,$$

and, for absolute constants $c, C > 0$,

$$\frac{c}{\log \frac{1}{\delta}} \le \mathrm{Fav}(S_\delta) \le \frac{C}{\log \frac{1}{\delta}} \qquad (0 < \delta < \frac{1}{2}).$$

Its length measure $\mu = \mathcal{H}^1|_S$ has uniform coordinate marginals.

Corollary 2 (KC2)**.** *There is an absolute constant C such that, for every $n \ge 2$,*

$$\frac{1}{H_n} \le \alpha_n \le \frac{C}{\log n}, \qquad H_n = \sum_{j=1}^{n} \frac{1}{j}.$$

The marginal condition means that $\mu(B \times [0, 1]) = \mu([0, 1] \times B) = |B|$ for Borel $B \subset [0, 1]$. A probability measure with these marginals is a *permuton* [4]. In our construction, the marginal condition follows from length 1.

Chang, Shmerkin and Suomala [1] obtain sharp Favard decay for random Cantor sets. Csörnyei and Suomala [2] characterize the measures which can occur as one projection of a purely unrectifiable length measure. We require both marginals, together with the quantitative bound. The fixed-base construction is too restrictive: for nested L-adic permutation patterns, [1, Theorem A.1] gives Favard length $\gtrsim_L \frac{\log j}{j}$ at scale L^{-j}. We instead allow the widths of the squares to vary. Throughout, c, C are positive absolute constants, allowed to change.

1. The matching reduction

Length 1 already forces the marginal condition. Indeed, suppose $p_x(S) = p_y(S) = [0, 1]$ and $\mathcal{H}^1(S) = 1$, and put $\mu = \mathcal{H}^1|_S$. For either coordinate projection p and Borel $B \subset [0, 1]$, the set $S \cap p^{-1}(B)$ projects onto B. Since p is 1-Lipschitz, $\mu(p^{-1}(B)) \ge |B|$. The same inequality for the complement, together with total mass 1, forces equality.

Put $\mathcal{A}(S) = \int_0^1 |\pi_t(S)|\, dt$. The angular change of variables $t = \frac{\sin\theta}{\cos\theta + \sin\theta}$ gives

$$\mathcal{A}(S) = \int_0^{\frac{\pi}{2}} \frac{|p_\theta(S)|}{(\cos\theta + \sin\theta)^3}\, d\theta \le \mathrm{Fav}(S). \tag{1}$$

[1] KC2 abbreviates "Kaczynski's Conjecture 2," following its numbering in Kuklinski [5, § 3].

Let μ have uniform coordinate marginals and be supported on S. Partition $[0,1]$ into n intervals I_i of length $\frac{1}{n}$, half-open except for the last, which includes 1. The matrix $a_{ij} = n\mu(I_i \times I_j)$ has every row sum and column sum equal to 1. Any k rows have total mass k and therefore meet at least k columns, since each column has mass 1. This is Hall's condition, so the matching theorem [3] supplies a permutation σ with $a_{i,\sigma(i)} > 0$ for all i. Every selected cell meets S, so, closing the cells,

$$P_\sigma := \bigcup_{i=1}^{n} \overline{I}_i \times \overline{I}_{\sigma(i)} \subset S_{\frac{1}{n}}. \tag{2}$$

The slice of the associated strip union at height t is $\pi_t(P_\sigma)$. Thus the upper bound in Theorem 1 implies the corollary through

$$\alpha_n \le \mathcal{A}(P_\sigma) \le \mathcal{A}(S_{\frac{1}{n}}) \le \mathrm{Fav}(S_{\frac{1}{n}}) \le \frac{C}{\log n}. \tag{3}$$

For $n = 2$, increase C if necessary.

The lower bound is a pairwise overlap count. In any permutation-strip arrangement of area A, let Z count the strips covering a point. Then $\int Z = 1$ and Cauchy–Schwarz gives $1 \le A \int Z^2$. Two crossing strips with endpoint index gaps $p, q \ge 1$ have intersection area

$$\frac{1}{n(p+q)} \le \frac{1}{4n}\left(\frac{1}{p} + \frac{1}{q}\right).$$

Their signed displacement is $\frac{p-(p+q)t}{n}$, so the overlap length has a triangular graph of height $\frac{1}{n}$ and base $\frac{2}{p+q}$. Noncrossing pairs have no positive-area intersection. On either edge there are $n-d$ pairs of index gap d. Counting each pair twice in Z^2 therefore gives

$$\int Z^2 \le 1 + \frac{1}{n}\sum_{d=1}^{n-1} \frac{n-d}{d} = H_n, \qquad A \ge \frac{1}{H_n}. \tag{4}$$

For any uniform-marginal measure, take $n = \lceil \frac{1}{\delta} \rceil$ in (2). Since $\frac{1}{n} \le \delta$, equations (1) and (4) give $\mathrm{Fav}(S_\delta) \ge \mathcal{A}(P_\sigma) \ge \frac{1}{H_n} \ge \frac{c}{\log \frac{1}{\delta}}$. Thus the Favard lower bound holds for every compact set carrying a probability measure with uniform marginals. We now construct a set attaining it with length 1.

2. Finite patterns

A *pattern* is a set $P = \bigcup_{i=1}^{N} (I_i \times J_i)$, where each family of closed intervals (I_i) and (J_i) covers $[0,1]$ with disjoint interiors, and $|I_i| = |J_i| = w_i > 0$. Write $N(P)$ for its number of squares. The sum of their sides is 1.

A red–blue shuffle preserves both interval partitions while reducing projection length. We also control the number of squares: thickening N squares by δ costs at most $CN\delta$ in Favard length.

Lemma 1 (Finite patterns)**.** *For every integer $k \ge 1$ there is a pattern P such that*

$$\mathrm{Fav}(P) \le \frac{C}{k}, \qquad \log N(P) \le Ck.$$

Proof. Each shuffle will give a quadratic expected saving in Favard length at bounded expected cost in $\log N$. Given a pattern P, write $E_t = \pi_t(P)$, $\ell(t) = |E_t|$ and $L(P) = \int_0^{\frac{1}{2}} \ell(t)\,dt$. We first estimate the overlap for $0 \le t \le \frac{1}{2}$.

Choose a width $0 < w < 1$ and partition a unit interval into $m = \lceil \frac{1}{w} \rceil$ pieces of width w, with the last possibly shorter. Take red and blue copies of this partition, and place them on coordinate intervals of length 2 in the orders

$$\begin{aligned} &x\text{-axis}: \quad R_1, B_1, R_2, B_2, \ldots, R_m, B_m, \\ &y\text{-axis}: \quad B_1, B_2, \ldots, B_m, R_1, R_2, \ldots, R_m. \end{aligned} \tag{5}$$

Equal labels determine squares. Put a homothetic copy of P inside each square, then divide both coordinates by 2. The resulting pattern P_w has $MN(P)$ squares, where $M = 2m$. Its copy ratios sum to 1, so subadditivity gives $|\pi(P_w)| \le |\pi(P)|$ for every linear projection π.

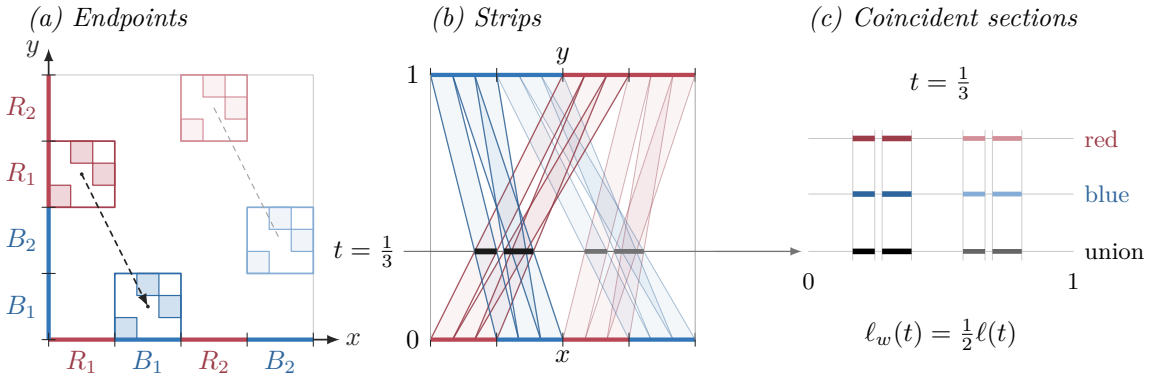


FIGURE 1. The shuffle for $w = \frac{1}{2}$. Each labeled square contains the same three-square pattern; the first red–blue pair is emphasized. Its translation follows a fiber of $\pi_{\frac{1}{3}}$, so the paired copies have identical sections at $t = \frac{1}{3}$. In (c), the red and blue sections are drawn separately; their union is the section in (b).

Within either color, the enclosing squares occur in the same order on both axes, so their π_t-projections have disjoint interiors. Each color contributes exactly $\frac{\ell(t)}{2}$. Thus, writing $\ell_w(t) = |\pi_t(P_w)|$, we have $\ell_w(t) \le \ell(t)$, and the difference is the overlap between the colors.

For a full-width pair, blue starts w to the right of red on the x-axis and one unit to its left on the y-axis, before scaling. The projected displacement, in units of their common side, is $d_w(t) = (1-t) - \frac{t}{w}$. The full-width pieces occupy at least half of each unit partition. Their red squares therefore have total side length at least $\frac{1}{4}$ after scaling. Adding the overlaps with their blue partners gives

$$\ell(t) - \ell_w(t) \ge \frac{1}{4} \left| E_t \cap (E_t + d_w(t)) \right|. \tag{6}$$

There is no overcounting because the red projections have disjoint interiors.

Averaging over translations makes the overlap depend only on length: for measurable $E \subset [0,1]$, Fubini gives [6]

$$\int_{\mathbb{R}} |E \cap (E+s)|\, ds = |E|^2. \tag{7}$$

The integral counts the pairs in $E \times E$. Its integrand is even and vanishes outside $[-1,1]$.

For $0 < t \le \frac{1}{2}$, a negative shift $d_w(t) = -s$, $0 \le s \le 1$, requires $w = \frac{t}{1-t+s} \asymp t$. Small heights need small widths, so we sample $w = U^2$, with U uniform on $(0,1)$. Then

$$U = \sqrt{\frac{t}{1-t+s}}, \qquad |dU| = \frac{\sqrt{t}}{2(1-t+s)^{\frac{3}{2}}}\, ds \gtrsim \sqrt{t}\, ds.$$

These values lie in $(0,1]$. Evenness and (7) give $\mathbb{E}_w[\ell(t) - \ell_w(t)] \gtrsim \sqrt{t}\, \ell(t)^2$. The square law gives an integrable reciprocal weight: $\int_0^{\frac{1}{2}} \frac{dt}{\sqrt{t}} < \infty$. Weighted Cauchy–Schwarz gives

$$\mathbb{E}_w[\mathcal{A}(P) - \mathcal{A}(P_w)] \gtrsim \int_0^{\frac{1}{2}} \sqrt{t}\, \ell(t)^2\, dt \gtrsim \left(\int_0^{\frac{1}{2}} \ell(t)\, dt \right)^2 = L(P)^2. \tag{8}$$

Since the weight in (1) is at most 1, the saving in Favard length is at least the saving in $\mathcal{A}$.

To cover all directions, let $T(x,y) = (y,x)$ and $R(x,y) = (x, 1-y)$, and first choose G uniformly from $\{I, T, R, T \circ R\}$. The map T exchanges t and $1-t$, while R supplies the other angular quadrant. Hence

$$\sum_G L(GP) = \mathcal{A}(P) + \mathcal{A}(RP) \asymp \mathrm{Fav}(P). \tag{9}$$

Apply the shuffle to GP. Favard length is invariant under G; averaging (8) and using Cauchy–Schwarz in (9) gives the quadratic saving

$$\mathbb{E}[\mathrm{Fav}(P) - \mathrm{Fav}((GP)_w)] \ge c\, \mathrm{Fav}(P)^2.$$

Each shuffle multiplies the number of squares by M, so its contribution to $\log N$ is $\log M$. Since $M = 2\lceil \frac{1}{w} \rceil \le \frac{4}{w} = \frac{4}{U^2}$, $\mathbb{E}\log M \le \log 4 + 2\int_0^1 \log \frac{1}{u}\, du < \infty$.

Starting with the unit square, use fresh independent choices of G, U at each of k steps. Let F_k, N_k be the resulting Favard length and number of squares, and put $f_k = \mathbb{E}F_k$. Conditional expectation and Jensen's inequality yield

$$f_{k+1} \le f_k - cf_k^2, \qquad f_k \le \frac{4}{1+4ck}, \qquad \mathbb{E}\log N_k \le Ck.$$

Here $f_0 = 4$, and $\frac{1}{f_{k+1}} \ge \frac{1}{f_k} + c$ proves the middle bound. To obtain both bounds for the same pattern, use their normalized sum: $\mathbb{E}[kF_k + \frac{\log N_k}{k}] \le C$. Some realization has this sum at most C; since both terms are nonnegative, it satisfies both estimates in the lemma. □

In particular, choosing k of order 2^j supplies patterns P_j with

$$\mathrm{Fav}(P_j) \le 2^{-j}, \qquad \log N(P_j) \le C2^j. \tag{10}$$

3. The limiting set

To nest patterns, insert a copy of Q into each square of $P = \bigcup_i (v_i + w_i[0,1]^2)$:

$$P \star Q = \bigcup_i (v_i + w_i Q).$$

The result is again a pattern, with $N(P)N(Q)$ squares. Moreover, $P \star Q \subset P$, while $\sum_i w_i = 1$ gives, for every linear projection π,

$$|\pi(P \star Q)| \le \min\{|\pi(P)|, |\pi(Q)|\}. \tag{11}$$

The squares of a pattern have total diameter $\sqrt{2}$. To obtain length 1, we group smaller squares inside copies of a pattern whose diameter is close to 1. For even m, let B_m be the equal-grid pattern given by the cyclic half-shift $i \mapsto i + \frac{m}{2} \pmod m$. It lies in the diamond

$$|x - \frac{1}{2}| + |y - \frac{1}{2}| \le \frac{1}{2} + \frac{1}{m}.$$

The diamond has diameter $1 + \frac{2}{m}$, so

$$\operatorname{diam} B_m \le 1 + \frac{2}{m}. \tag{12}$$

Thus insertion into B_m preserves the projection bound and gives a cover of diameter tending to 1.

(a) The half-shift *(b) Insertion into a pattern*

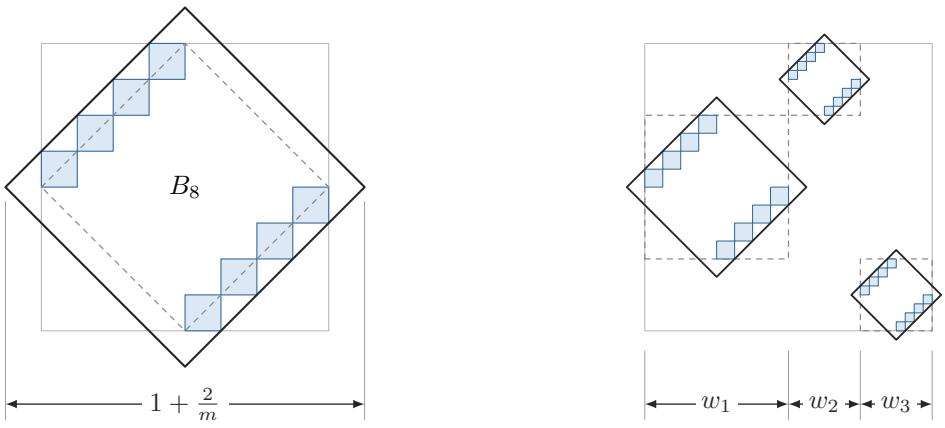


FIGURE 2. The half-shift B_8 and its insertion into three squares of widths $w_1 = \frac{1}{2}$, $w_2 = w_3 = \frac{1}{4}$. The dashed diamond has diameter 1. The solid diamonds enclose the copies; on the right their total diameter is $\left(1 + \frac{2}{m}\right)\sum_i w_i = 1 + \frac{2}{m}$, with $m = 8$.

Set $Q_j = B_{2j} \star P_j$. Form the nested patterns $T_0 = [0,1]^2$ and $T_j = T_{j-1} \star Q_j$. Equations (10)–(12) give

$$\mathrm{Fav}(T_j) \le C2^{-j}, \qquad \log N(T_j) \le C2^j. \tag{13}$$

For the second estimate, sum the logarithmic costs of the factors: $\sum_{i \le j} (\log(2i) + C2^i) \le C2^j$. The factor B_{2j} makes every child square at most half as wide as its parent, so the largest square side in T_j is at most 2^{-j}.

Let $S = \bigcap_j T_j$. The sets $v_i + w_i B_{2(j+1)}$, indexed by the squares $v_i + w_i[0,1]^2$ of T_j, cover S. Their diameters tend uniformly to zero, and their total diameter is at most $\sum_i w_i(1 + \frac{1}{j+1}) = 1 + \frac{1}{j+1}$. Thus $\mathcal{H}^1(S) \le 1$. Both coordinate projections of S are $[0,1]$, by nested compactness, so $\mathcal{H}^1(S) \ge 1$ as well.

It remains to check the rate between successive construction scales. Enlarging a union of N squares by $[-\delta, \delta]^2$ increases its Favard length by at most $8N\delta$: in direction θ, each projected interval gains $2\delta(|\cos\theta| + |\sin\theta|)$. For a fixed absolute $B > 0$, equation (13) gives

$$\mathrm{Fav}(S_\delta) \le C2^{-j} + 8e^{B2^j}\delta.$$

For small δ, choose the largest j with $e^{B2^j} \le \delta^{-\frac{1}{2}}$. This keeps the thickening error at most $8\sqrt{\delta}$, while $2^j \asymp \log \frac{1}{\delta}$. Thus

$$\mathrm{Fav}(S_\delta) \le \frac{C}{\log \frac{1}{\delta}} + 8\sqrt{\delta} \le \frac{C}{\log \frac{1}{\delta}}.$$

The remaining values of δ are covered by increasing C. Section 1 supplies the uniform marginals and the matching lower estimate, completing the proof. In particular, $\mathrm{Fav}(S) = 0$ and $\mathcal{H}^1(S) = 1$; the Besicovitch projection theorem implies that S is purely unrectifiable.

Conversely, KC2 supplies equal-grid patterns with $\mathcal{A}(P_j) \le 2^{-j}$ and $\log N(P_j) \le C2^j$: take $n_j = \lceil \exp(C_* 2^j) \rceil$. Equations (1) and (11) give $\mathrm{Fav}(P_j \star R(P_j)) \le C\mathcal{A}(P_j)$: the two factors control the two angular quadrants. Thus KC2 also gives (10), after shifting j by a fixed amount. The same half-shifts and nested insertions give Theorem 1. The theorem and KC2 are therefore equivalent.

Acknowledgment. The main geometric idea originated with the author. Some of the more intricate computational details were worked out with assistance from OpenAI's Astra.

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